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The sharp range of the bipanchromatic number of a hypergraph

Matias Ortiz[1,*]

[1] *New York University*

Abstract

We determine the exact range of the bipanchromatic number of a finite hypergraph with prescribed vertex count, panchromatic number, and minimum number of singleton classes among optimal panchromatic colorings. In the nontrivial regime, the possible values form an integer interval with explicit endpoints. Every value is attained by a uniform hypergraph whose edge size equals its panchromatic number and whose common-core size equals the prescribed singleton parameter. The proof uses a recoloring construction, balanced class sizes, and unit transfers between part sizes. Consequently, the pairing lower bound is best possible when only the two coloring parameters are fixed, and an exact vertex-count threshold forces the largest possible bipanchromatic number. The main theorem, including its uniform realization statement, has been formalized in Lean.

Keywords: hypergraph coloring; panchromatic coloring; bipanchromatic coloring; transversal; extremal hypergraph.

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1 Introduction

A panchromatic coloring partitions the vertices of a hypergraph into transversals. A bipanchromatic coloring additionally requires every class to contain at least two vertices. Panchromatic coloring is also studied as polychromatic coloring; see [1]. Lalou, Mbarek, Skender, and Togni used these parameters to study completely independent spanning trees in split graphs [2]. In their subsequent work on chordal graphs [3], they disproved the identity

$$\chi_p^2(H) = \chi_p(H) - \lceil \alpha(H)/2 \rceil ,$$

where $\alpha(H)$ is the minimum number of singleton classes among optimal panchromatic colorings. The right-hand side is still a lower bound, obtained by pairing singleton classes. They ask for sharper bounds involving $\chi_p(H)$ and $\alpha(H)$ [3, Section 5].

We determine the complete range when the vertex count is fixed as well. For each feasible triple (n, p, a) with $1 \le a < p$, the attainable bipanchromatic numbers form an explicit integer interval. Every value is realized by a p-uniform hypergraph with common core of size a. The proof combines recoloring, balanced class sizes, an exact construction, and unit transfers between its part sizes. As consequences, the pairing bound is best possible when only (p, a) are fixed, while specifying n yields the optimal refinement. The results concern hypergraph coloring; applications to spanning trees require the structural hypotheses of the relevant graph correspondence.

2 Definitions and the main theorem

Throughout, $H = (V, E)$ is a hypergraph with finite vertex set V, $|V| \ge 2$, and a nonempty family E of nonempty subsets of V. Isolated vertices are permitted. A *transversal* is a set meeting every edge of H. A *panchromatic coloring* is a partition of V into nonempty transversals. A *bipanchromatic coloring* is a panchromatic coloring in which every class has size at least two. This size requirement is global; it does not require two vertices of each color in each edge. We allow a one-class partition, so both coloring parameters are defined under these hypotheses.

The panchromatic and bipanchromatic numbers, denoted by $\chi_p(H)$ and $\chi_p^2(H)$, are the maximum numbers of classes in the respective colorings. The superscript 2 in χ_p^2 is part of the notation. For an optimal panchromatic coloring $\mathcal{P}$, let

$$u(\mathcal{P}) = |\{P \in \mathcal{P} : |P| = 1\}|, \qquad \alpha(H) = \min_{\substack{\mathcal{P} \text{ panchromatic} \\ |\mathcal{P}| = \chi_p(H)}} u(\mathcal{P}).$$

Thus α denotes a singleton parameter, not an independence number. Finiteness ensures that all the extrema in these definitions are attained. Write

$$n = |V|, \qquad p = \chi_p(H), \qquad a = \alpha(H), \qquad b = \chi_p^2(H).$$

*Corresponding author (mo2273@nyu.edu).

The *common core* is $\operatorname{core}(H) = \bigcap_{e \in E} e$. A singleton is a transversal precisely when its vertex belongs to the common core. In general, $\alpha(H)$ need not equal $|\operatorname{core}(H)|$. A hypergraph is *p-uniform* if every edge has size p. For a real number x, let $x_+ = \max\{x, 0\}$.

Theorem 2.1 (Sharp range)**.** *Let n, p, a be integers satisfying*

$$p \geq 2, \qquad 1 \leq a < p, \qquad n \geq 2p - a. \tag{2.1}$$

Define

$$q = p - a, \qquad s = \left\lfloor \frac{n-a}{q} \right\rfloor, \qquad r = n - a - qs, \qquad B_t = st + \min\{t, r\} \quad (0 \leq t \leq q).$$

In particular, $q \geq 1$, $s \geq 2$, and $0 \leq r < q$. Set

$$L(n, p, a) = p - \min_{0 \leq t \leq q} \left(t + \left\lceil \frac{(a - B_t)_+}{2} \right\rceil \right), \tag{2.2}$$

$$U(n, p, a) = \min\{p - 1, \lfloor n/2 \rfloor\}. \tag{2.3}$$

The set of attainable bipanchromatic numbers of hypergraphs H with $|V(H)| = n$, $\chi_p(H) = p$, and $\alpha(H) = a$ is exactly

$$\{L(n, p, a),\ L(n, p, a) + 1, \ldots, U(n, p, a)\}.$$

Every value in this interval is attained by a p-uniform hypergraph whose common core has exactly a vertices.

The parameter q is the number of nonsingleton classes in an optimal coloring attaining a. Their total size is $n - a$. The integers s and r are its quotient and remainder on division by q, and B_t is the sum of the t largest entries of the balanced vector with r entries $s + 1$ and $q - r$ entries s. All minima and maxima indexed by t in this paper are over integer values.

The condition $n \geq 2p - a$ is necessary by counting such a coloring. The construction in the proof shows that it is sufficient when $1 \leq a < p$. The boundary cases are recorded in Proposition 6.1.

3 Recoloring and balanced class sizes

Let $a, q \geq 1$ and let $d = (d_1, \ldots, d_q)$ be a vector of integers at least two. Arrange its entries in nonincreasing order and put

$$p = a + q, \qquad S_0(d) = 0, \qquad S_t(d) = \sum_{i=1}^{t} d_i \quad (1 \leq t \leq q).$$

The functions S_t are independent of how equal entries are ordered. Define

$$F_a(d) = \max_{0 \leq t \leq q} \left(p - t - \left\lceil \frac{(a - S_t(d))_+}{2} \right\rceil \right). \tag{3.1}$$

Lemma 3.1 (Dissolving color classes)**.** *Suppose a panchromatic coloring of H has $a \geq 1$ singleton classes and $q \geq 1$ nonsingleton classes, with size vector d. Then $\chi_p^2(H) \geq F_a(d)$.*

Proof. Fix $t \in \{0, \ldots, q\}$ and retain the $q - t$ nonsingleton classes other than the t largest. Dissolve those t classes into a pool of $S_t = S_t(d)$ vertices. Let $m = \min\{a, S_t\}$. Pair m singleton vertices with distinct vertices from the pool, and then pair the remaining $a - m$ singleton vertices among themselves as far as possible. Every resulting pair is a transversal because it contains a vertex whose singleton was a transversal. The retained classes remain transversals. All these sets are pairwise disjoint and have size at least two, and their number is

$$\begin{aligned}(q - t) + m + \left\lfloor \frac{a - m}{2} \right\rfloor &= p - t - \left\lceil \frac{a - m}{2} \right\rceil \\ &= p - t - \left\lceil \frac{(a - S_t)_+}{2} \right\rceil.\end{aligned}$$

At least one set has been constructed. Indeed, if $t < q$, a retained class exists; if $t = q$, then $S_q \geq 2q \geq 2$ and $m \geq 1$. Add every unused vertex to any one of the constructed sets. This preserves the transversal property and the minimum class size, and gives a partition of V. Taking the maximum over t proves the assertion. □

Lemma 3.2 (Balanced largest-part sums)**.** *Let $d_1 \geq \cdots \geq d_q$ be integers with total $qs + r$, where $q \geq 1$ and $0 \leq r < q$. Then*

$$S_t(d) \geq st + \min\{t, r\} \qquad (0 \leq t \leq q). \tag{3.2}$$

Equality holds for every t when d has r entries $s + 1$ and $q - r$ entries s.

Proof. The case $t = 0$ is immediate. Suppose $t \geq 1$ and the asserted inequality fails. If $t \leq r$, integrality gives $S_t \leq t(s+1)-1$, hence $d_t \leq s$. All later entries are also at most s, and consequently

$$\sum_{i=1}^{q} d_i \leq S_t + (q-t)s \leq qs + t - 1 < qs + r,$$

a contradiction. If $t > r$, then $S_t \leq st + r - 1 < t(s+1)$, so again $d_t \leq s$. It follows that

$$\sum_{i=1}^{q} d_i \leq S_t + (q-t)s \leq qs + r - 1,$$

also a contradiction. The equality assertion follows by summing the largest entries of the specified balanced vector. □

4 An exact construction

Lemma 4.1 (Hypergraph associated with a size vector)**.** *Let $a, q \geq 1$ and $d_1 \geq \cdots \geq d_q \geq 2$ be integers. Choose pairwise disjoint sets $C, A_1, \ldots, A_q$ with $|C| = a$ and $|A_i| = d_i$. Define $H(a;d) = (V, E)$ by*

$$V = C \,\dot\cup\, A_1 \,\dot\cup\, \cdots \,\dot\cup\, A_q,$$
$$E = \big\{C \cup \{v_1, \ldots, v_q\} : v_i \in A_i \text{ for every } i\big\}.$$

Here $\dot\cup$ denotes disjoint union. Then $H(a;d)$ is $(a+q)$-uniform and satisfies

$$|V| = a + \sum_{i=1}^{q} d_i, \qquad \chi_p(H(a;d)) = a + q, \qquad \alpha(H(a;d)) = a, \qquad \operatorname{core}(H(a;d)) = C. \tag{4.1}$$

Moreover,

$$\chi_p^2(H(a;d)) = F_a(d). \tag{4.2}$$

Proof. The edge family is nonempty, and every edge has $p := a + q$ vertices. Every edge contains C. If $v \in A_i$, an edge can omit v by choosing another vertex from A_i, since $d_i \geq 2$. Thus the common core is exactly C.

The a singleton subsets of C, together with $A_1, \ldots, A_q$, form a panchromatic coloring with p classes. Every class of any panchromatic coloring must meet a fixed edge of size p, so no such coloring has more than p classes. Hence $\chi_p(H(a;d)) = p$.

In a panchromatic coloring with p classes, each edge contains exactly one vertex from each class. Fix $u \in C$. For every $v \neq u$, some edge contains both u and v. This is immediate if $v \in C$, and otherwise follows by choosing v from its part. Therefore u and v cannot belong to the same class. Every vertex of C is a singleton class in every optimal coloring. No vertex outside C is a singleton transversal, because some edge omits it. Thus every optimal coloring has exactly a singleton classes, proving $\alpha(H(a;d)) = a$.

A set $T \subseteq V \setminus C$ is a transversal if and only if it contains an entire part A_i. One implication follows because every edge meets every part. For the other, if $A_i \nsubseteq T$ for all i, choose $v_i \in A_i \setminus T$ for each i. The edge $C \cup \{v_1, \ldots, v_q\}$ then avoids T.

Consider any bipanchromatic coloring. Let k be the number of its classes avoiding C and x the number meeting C. Each of the k classes contains an entire part. Since the classes are disjoint, different classes must contain different parts. Hence $k \leq q$, and their union has size at least the sum of the k smallest entries of d. Set $t = q - k$. This sum is $\sum_i d_i - S_t(d)$, leaving at most $a + S_t(d)$ vertices for the x other classes. These classes each contain a core vertex and at least two vertices in total. Therefore

$$\begin{aligned} x &\leq \min\left\{a, \left\lfloor \frac{a + S_t(d)}{2} \right\rfloor\right\} \\ &= a - \left\lceil \frac{(a - S_t(d))_+}{2} \right\rceil. \end{aligned}$$

The coloring consequently has at most $p - t - \lceil (a - S_t(d))_+/2 \rceil \leq F_a(d)$ classes. Conversely, the natural coloring displayed above has the class sizes required by Lemma 3.1, which gives $\chi_p^2(H(a;d)) \geq F_a(d)$. This proves (4.2). □

5 Unit transfers and the proof of the main theorem

Lemma 5.1 (Unit transfer)**.** *Let d be a vector of q integers at least two. Increase a largest entry by one and decrease a different entry greater than two by one, obtaining d'. Then*

$$F_a(d) \leq F_a(d') \leq F_a(d) + 1.$$

Proof. Let i be the receiving index and j the donating index, using the indices before sorting. For each t, the sum $S_t(d)$ is the largest sum over all t-element sets of indices. If $t \geq 1$, a maximizing index set can be chosen to contain i, since d_i is a largest entry. Its sum increases by zero if it also contains j, and by one otherwise. Thus $S_t(d') \geq S_t(d)$. The sum on any fixed index set increases by at most one, so $S_t(d') \leq S_t(d) + 1$. Both sums are zero for $t = 0$.

If nonnegative integers S, S' satisfy $S \leq S' \leq S+1$, then $\lceil (a-S)_+/2 \rceil - \lceil (a-S')_+/2 \rceil$ is either zero or one. Consequently each term in the maximum defining $F_a(d')$ lies between the corresponding term for $F_a(d)$ and that term plus one. Taking maxima gives the result. □

Proof of Theorem 2.1. We first establish the bounds for an arbitrary hypergraph with the prescribed parameters. Choose an optimal panchromatic coloring attaining a, and let $d_1 \geq \cdots \geq d_q \geq 2$ be its nonsingleton class sizes. Their sum is $n - a = qs + r$. Lemma 3.2 gives $S_t(d) \geq B_t$. Each term in (3.1) is nondecreasing in $S_t(d)$, so Lemma 3.1 yields $\chi_p^2(H) \geq L(n,p,a)$.

Every bipanchromatic class has size at least two, whence $\chi_p^2(H) \leq \lfloor n/2 \rfloor$. Also $\chi_p^2(H) \leq p$. Equality with p would give an optimal panchromatic coloring with no singleton classes, contradicting $a > 0$. Thus $\chi_p^2(H) \leq U(n,p,a)$.

For the lower endpoint, apply Lemma 4.1 to the balanced vector

$$d^{\mathrm{bal}} = (\underbrace{s+1, \ldots, s+1}_{r \text{ entries}}, \underbrace{s, \ldots, s}_{q-r \text{ entries}}).$$

It has total $n - a$ and satisfies $S_t(d^{\mathrm{bal}}) = B_t$ for every t. Its associated hypergraph therefore has the required parameters and bipanchromatic number $L(n,p,a)$.

For the upper endpoint, let $D = n - a - 2(q-1) \geq 2$ and consider

$$d^{\mathrm{con}} = (D, \underbrace{2, \ldots, 2}_{q-1 \text{ entries}}).$$

For $t \geq 1$, its largest-part sum is $D + 2(t-1)$. The minimum, over $1 \leq t \leq q$, of the associated losses

$$t + \left\lceil \frac{(a - D - 2(t-1))_+}{2} \right\rceil$$

is $1 + \lceil (a-D)_+/2 \rceil$, attained at $t = 1$. Indeed, if $a \leq D$, each loss is at least $t \geq 1$ and the first loss is one. If $a > D$, removing the positive part gives the lower bound

$$t + \left\lceil \frac{a - D - 2(t-1)}{2} \right\rceil = 1 + \left\lceil \frac{a-D}{2} \right\rceil,$$

again attained at $t = 1$. This minimum is no greater than the $t = 0$ loss $\lceil a/2 \rceil$: for $a \leq D$ this follows from $a \geq 1$, and for $a > D$ it follows from $D \geq 2$. Thus

$$\begin{aligned} F_a(d^{\mathrm{con}}) &= p - 1 - \left\lceil \frac{(a-D)_+}{2} \right\rceil \\ &= \min\{p-1, \lfloor n/2 \rfloor\} = U(n,p,a). \end{aligned}$$

For the second equality, use $a - D = 2p - n - 2$. If this number is nonpositive, both sides equal $p - 1$; otherwise the half-ceiling subtracted from $p - 1$ gives $\lfloor n/2 \rfloor$. Lemma 4.1 realizes this value.

Finally, start at d^{bal} and fix one of its largest entries as receiver. Whenever another entry exceeds two, transfer one unit from it to the receiver. The receiver remains largest. The nonnegative integer

$$\sum_{j \neq i} (d_j - 2),$$

where i is the receiver, decreases by one at each step, so the process terminates at d^{con}. When $q = 1$, the two vectors already coincide and no transfer is needed.

All vectors along this finite path have q entries at least two and total $n - a$. By Lemma 4.1, each gives a p-uniform hypergraph with the prescribed parameters and common core of size a. By Lemma 5.1, consecutive bipanchromatic numbers increase by either zero or one. They start at $L(n,p,a)$ and end at $U(n,p,a)$. For any integer strictly between the endpoints, the first value along the path that is at least that integer must equal it. Hence every integer in the interval is attained, as claimed. □

6 Consequences and boundary cases

Proposition 6.1 (Boundary parameters)**.** *Under the standing assumptions,* $0 \le a \le p$ *and* $p \ge 1$*. The cases outside* $1 \le a < p$ *are as follows.*

1. *If* $a = 0$*, then* $n \ge 2p$ *and* $b = p$*. Conversely, every pair* $p \ge 1$*,* $n \ge 2p$ *occurs with* $a = 0$ *and* $b = p$*.*
2. *If* $a = p$*, then* $n = p \ge 2$*,* $E = \{V\}$*, and* $b = \lfloor p/2 \rfloor$*. Conversely, these parameters occur for every* $p \ge 2$*.*

In particular, if $p = 1$ *then* $a = 0$ *and* $b = 1$*.*

Proof. The elementary inequalities follow from the definitions. If $a = 0$, an optimal panchromatic coloring has all classes of size at least two. It is bipanchromatic, proving $b = p$ and $n \ge 2p$. For the converse, partition n vertices into p parts of size at least two and take as edges all sets containing one vertex from each part. The parts give a bipanchromatic p-coloring, while every edge has size p and therefore bounds the panchromatic number by p.

If $a = p$, an optimal coloring attaining a consists of p singleton classes, which cover all vertices. Thus $n = p$, and every edge contains all vertices. Conversely, when $E = \{V\}$, the panchromatic number is n, its only optimal partition consists of singletons, and partitioning into pairs with a possible remaining vertex attains $\lfloor n/2 \rfloor$ bipanchromatic classes. Finally, a one-class partition of a set with at least two vertices has no singleton class, giving the assertion for $p = 1$. □

Corollary 6.1 (Optimality of the pairing guarantee)**.** *Every hypergraph under the standing assumptions satisfies*

$$\chi_p^2(H) \ge \chi_p(H) - \lceil \alpha(H)/2 \rceil . \tag{6.1}$$

For every $p \ge 2$ *and* $1 \le a < p$*, equality occurs with* $\chi_p(H) = p$ *and* $\alpha(H) = a$*. Thus the lower bound in* (6.1) *cannot be increased at any such pair* (p, a) *without additional hypotheses or parameters.*

Proof. In the nontrivial regime, take $t = 0$ in Lemma 3.1. The boundary cases follow from Proposition 6.1. To attain equality for fixed p, a, set $q = p - a$ and use Lemma 4.1 with all q part sizes equal to two. The resulting hypergraph has $n = a + 2q = 2p - a$ vertices, and hence

$$\chi_p^2(H) \le \left\lfloor \frac{2p - a}{2} \right\rfloor = p - \lceil a/2 \rceil .$$

Together with (6.1), this proves equality. □

Corollary 6.2 (Threshold for the upper value)**.** *Fix a feasible triple* (n, p, a) *satisfying* (2.1)*. Every hypergraph with these parameters has* $\chi_p^2(H) = p - 1$ *if and only if*

$$n > a + (p - a)(a - 1). \tag{6.2}$$

Proof. If (6.2) holds, the $q = p - a$ nonsingleton classes of an optimal coloring attaining a have total $n - a > q(a - 1)$. At least one has size at least a. Dissolving that class supplies a distinct partner for every singleton class, giving $p - 1$ bipanchromatic classes. The upper bound $b \le p - 1$ then proves equality.

Conversely, suppose $n - a \le q(a - 1)$. Since $n - a \ge 2q$, this implies $a \ge 3$. In the balanced construction, the largest part has size $B_1 = \lceil (n - a)/q \rceil \le a - 1$. The $t = 0$ loss in (2.2) is at least two. The $t = 1$ loss is at least two because $B_1 < a$, and each $t \ge 2$ loss is at least two as well. This construction therefore has $b = L(n, p, a) \le p - 2$, proving the converse. □

Corollary 6.3 (Maximum additive improvement over pairing)**.** *For fixed* $p \ge 2$ *and* $1 \le a < p$*, the largest possible value of*

$$\chi_p^2(H) - \big(p - \lceil a/2 \rceil\big), \qquad \chi_p(H) = p, \quad \alpha(H) = a,$$

with the number of vertices unrestricted, is $\lceil a/2 \rceil - 1$*.*

Proof. The upper bound follows from $\chi_p^2(H) \le p - 1$. For any $n \ge \max\{2p - a, 2p - 2\}$, Theorem 2.1 realizes $U(n, p, a) = p - 1$, attaining the asserted difference. □

For illustration, Table 6.1 fixes $p = 10$ and $a = 6$. The pairing guarantee is always seven, whereas the exact lower endpoint can be eight or nine when the vertex count is specified.

For $(n, p, a) = (18, 10, 6)$, one has $q = 4$, $s = 3$, $r = 0$, and $(B_0, \ldots, B_4) = (0, 3, 6, 9, 12)$. The losses in (2.2) are $(3, 3, 2, 3, 4)$, giving $L = 8$ and $U = 9$. The transfer path

$$(3, 3, 3, 3) \longrightarrow (4, 3, 3, 2) \longrightarrow (5, 3, 2, 2) \longrightarrow (6, 2, 2, 2)$$

has exact bipanchromatic values $8, 8, 8, 9$ under the construction of Lemma 4.1.

The endpoints can be evaluated using $O(p - a)$ arithmetic operations once n, p, a are given. This computation does not provide an algorithm for finding p or a from an arbitrary hypergraph. Likewise, the realization construction is an extremal existence result: its explicit edge family has $\prod_i d_i$ members and may be large.

n	Balanced part sizes	$L(n, 10, 6)$	$U(n, 10, 6)$
14	$(2, 2, 2, 2)$	7	7
16	$(3, 3, 2, 2)$	8	8
18	$(3, 3, 3, 3)$	8	9
24	$(5, 5, 4, 4)$	8	9
27	$(6, 5, 5, 5)$	9	9

Table 6.1: Sharp intervals with panchromatic number ten and singleton parameter six.

7 Formal verification

Theorem 2.1, including the uniform witnesses with exact common-core size, is formalized in Lean 4.34.1 with mathlib v4.34.1. The accompanying archive `sharp_range_lean_certificate.zip` contains the proof sources, pinned dependency manifest, reproduction instructions, and verification records. The declarations `Bipanchromatic.sharp_range` and `Bipanchromatic.sharp_range_witness` in `SharpRange/Full.lean` prove the interval theorem and the stronger realization statement, respectively. The formal parameters are natural numbers satisfying $p \geq 2$, $1 \leq a < p$, and $2p \leq n + a$; the extrema and class-size conditions agree with Section 2.

Verification completed on October 9, 2026. The project builds successfully, and all 116 public proved declarations have axiom dependencies contained in `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx`, custom axioms, or native-evaluation axioms. A fresh replay of the final module and its imports through Lean's kernel also passed; this uses the same kernel, not an independent verifier [4]. The archive's `verify.py` reproduces these checks. The boundary cases and consequences in Section 6 are proved in the text and are not separately claimed as formalized.

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